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School of Architecture and Civil Engineering
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EDUCATION

■ Postdoc. Geotechnical Engineering, (City University of Hong Kong)	2026.04—Present
■ Research Assistant. Geotechnical Engineering, (City University of Hong Kong)	2025.11—2026.04
■ Ph.D. Geotechnical Engineering, (City University of Hong Kong, GPA: 3.85/4.0)	2021.09—2025.11
■ M.S., Geotechnical Engineering, (Hunan University, GPA: 85.3/100 (2%))	2015.09—2018.06
■ B.E., Geotechnical Engineering, (Hunan University, GPA: 3.27/4 (5%))	2011.09—2015.06

RESEARCH EXPERIENCE

- **Project name: “Development of an X-Ray Microtomography Method for Full-Field Discrete Particle Tracking in Sand Specimens” (Funding: General Research Fund, Hong Kong, Num. 11204224, HKD 1.13M)**
 - Role: Main member, CityU HK, 2025-present
 - Designed and conducted in-situ CT tests on calcareous sands (CS), highly and completely decomposed granite (HDG/CDG) particles.
 - Developed a novel image-processing framework for morphologically complex and highly fragmented particles, including CS, HDG, and CDG.
 - Developed a fast and robust full-field tracking method for multiple particle and fragment types (e.g., bead, Leighton Buzzard sand (LBS), HDG, CDG, CS).
- **Project name: “Development of A Hybrid Feature-Aided Volumetric Digital Image Correlation Method for Fine-Grained Soil Mixtures” (Funding: General Research Fund, Hong Kong, Num. 11207321, HKD 0.91M)**
 - Role: Member, CityU HK, 2022-2026
 - Developed a robust particle-segmentation method for noisy low-dose CT images.
 - Developed new imaging processing techniques to decrease the noise of CT images obtained from CU tests.
- **Project name: “Micromechanical Investigation of Sands Using Machine Learning Methods” (Funding: General Research Fund, Hong Kong, Num. 11201020, HKD 0.87M)**
 - Role: Member, CityU HK, 2021-2024
 - Designed and conducted in-situ CT tests on glass beads and quartz particles.
 - Processed raw CT images acquired during triaxial tests on beads and LBS particles
 - Recognized and tracked individual LBS particles using PointNet and PointNetLK networks.
- **Project name: “The nature of crushing behavior of calcareous sand and the micromechanical mechanism of microbial solidification” (Funding: NSFC-SRI, R-IND1011, RMB 0.60M)**
 - Role: Member, CityU HK, 2021-2023
 - Developed a new image-processing technique for accurate segmentation of calcareous sand particles.
 - Developed a novel particle tracking method for CS particles with severe particle breakage.
- **Project name: “Tackling Particle Morphology in the Micromechanical Study of Crushable Sands: A Combined Experimental, Theoretical and Numerical Approach” (Funding: General Research Fund, Hong Kong, Num. 11272916, HKD 0.68M)**
 - Role: Member, CityU HK, 2017-2020
 - Simulated particle breakage events in triaxial tests using PFC3D with the particle-replacement approach, in which virtual particle morphologies were generated from real LBS shape characteristics.

JOURNAL PAPERS

- **Zhu, Z.**, Lu, J., Huang, S., & Wang, J., (2026). A hybrid YOLOv26–SAM2 framework for accurate segmentation and reconstruction of calcareous sand particles. *Acta Geotechnica*. (Under Review)
- **Zhu, Z.**, Wang, J., & Zhao, B., (08/2026). X-ray micro-computed tomography investigation of particle breakage of highly decomposed granite aided by full-field particle tracking. *Journal of Geotechnical and Geoenvironmental Engineering*, 10.1061/JGGEFK/GTENG-13730.
- **Zhu, Z.**, & Wang, J. (01/2025). Matching of crushed highly decomposed granite particles using 3D SHOT descriptors. *Géotechnique*, 75(1), 16-28.
- **Zhu, Z.**, Wang, J., & Yu, H. (10/2024). Tracking the movement of quartz sand particles with neural networks. *Computers and Geotechnics*, 174, 106666.
- **Zhu, Z.**, Wang, J., & Wu, M. (01/2023). Pattern recognition of quartz sand particles with PointConv network. *Computers and Geotechnics*, 153, 105061.
- **Zhu, Z.**, Wang, J., & Wu, M. (08/2022). DEM simulation of particle crushing in a triaxial test considering the influence of particle morphology and coordination number. *Computers and Geotechnics*, 148, 104769.
- **Zhu, Z.**, Cheng, Q. Q., Chen, Y. Y., Chen, E. J., & Su, Y. (2021). Statistically Quantifying the Uncertainty in Strength of Marine Clay Stabilized by a Type of Portland-Fly Ash Cement. *Advances in Civil Engineering*, 2021(1), 6654802.
- Xia, Z., Wang, J., **Zhu, Z.**, Hu, W., Tracy, S., & Zhao, B. (2024). Progressive internal erosion during triaxial shearing: An X-ray micro-computed tomography study. *Géotechnique*, 1-12.
- Xiong, W., Wang, J., & **Zhu, Z.** (2025). Macro-micro mechanics of Fujian River sand under triaxial shear using X-ray micro-tomography. *Canadian Geotechnical Journal*, 62, 1-15.
- Yong Liu, Wengang Zhang, Lei Zhang, **Zhiren Zhu**, Jun Hu, Hong Wei, *Probabilistic Stability Analysis of Undrained Slopes by 3D Random Fields and Finite Element Methods*, Geoscience Frontiers. Vol 9 (6), 2018, 1657-1664. (**Highly Cited Paper**)
- Chen, G., **Zhu, Z.**, & Hu, J. (2021). Simulation of response spectrum-compatible ground motions using wavelet-based multi-resolution analysis. *Measurement and Control*, 54(5-6), 641-646.
- Yang, C., **Zhu, Z.**, & Xiao, Y. (2020). Bearing capacity of ring foundations on sand overlying clay. *Applied Sciences*, 10(13), 4675.
- Wu, G., Zhao, H., Zhao, M., & **Zhu, Z.** (2021). Stochastic analysis of dual tunnels in spatially random soil. *Computers and Geotechnics*, 129, 103861.
- Minghua Zhao, **Zhiren Zhu**, Minghua Huang, Chaoyi Yang, Zhuojun Xu, *Calculation Method of the Safe Thickness of Karst Cave Roof with the Influence of Rock-socketed Pile Foundation*, Rock and Soil Mechanics (Chinese), Vol 39 (11), 2018, 4201-4209.
- Xin Tan, Minghua Zhao, **Zhiren Zhu**, Yuxuan Jin, *Elastic Properties Calibration Approach for Discrete Element Method Model Based on Voronoi Tessellation Method*, Geotechnical and Geological Engineering, Vol 37 (3), June 2019, 2227-2236.

CONFERENCE PAPERS

- **Zhu Z.**, Jianfeng Wang. (2026, February). Study of Particle Breakage of Highly Decomposed Granite Particles Under a One-Dimensional Compression Test, The 17th International Conference of the International Association for Computer Methods and Advances in Geomechanics.
- **Zhu, Z.**, & Wang, J. (2025, April). Investigation of the Breakage Behaviour of Completely Decomposed Granite Under Shear Using X-ray Tomography. In *IOP Conference Series: Earth and Environmental Science* (Vol. 1480, No. 1, p. 012068). IOP Publishing.
- **Zhu, Z.**, & Wang, J. (2024, August). Tracking of Fragmented Particles with Neural Networks. In *International Conference on Information Technology in Geo-Engineering* (pp. 59-67). Cham: Springer Nature Switzerland.

AWARDS

- Institutional Research Tuition Scholarship (08/2023, 1%, Award, Competition, City University of Hong Kong)
- University Grants Committee Scholarship (09/2021, 1%, Award, Competition, City University of Hong Kong)
- National Scholarship (11/2017, 1%, Award, Competition, Hunan University)

- Third Prize in the Second Hunan Provincial University Students' Structural Design Competition (12/2013, 2%, Award, Competition, Hunan Provincial Department of Education)

TECHNICAL SKILLS

- Advanced: Avizo, FISH, PFC, PRITE, UDEC
- Proficient: Python, MATLAB, Dragonfly, ilastik